



LICEO SAVORE: Development of a Web-Based School Canteen Pre-Ordering and Pickup System with Hybrid Payment and No-Show Management to Improve Queue Management and Transaction Efficiency at Liceo de Cagayan University

MOCORRO, PRINCY LAIZA
DADO, JEFF BRIAN
MESIAS, AARON KHALIL
RESPONSO, ALTHEA NESSY
GUALIZA, CHRISTIAN RHEY
VELEZ, JANDY
RALLOMA, MARK CLARENCE
CAINGIN, DENNIS
LOQUIAS, KEN
ACIDO, ROGEL
ALBISO, MARY GRACE
JUMAHALI, JOHN DAVE

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MOCORRO, PRINCY LAIZA
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CHAPTER I INTRODUCTION

1.1 Background of the Study

School canteens provide students with access to food during the school day, particularly during limited breakfast and break periods. However, food-service transactions can become difficult to manage when many students attempt to purchase food within the same period. Previous studies have examined different factors affecting campus food-service operations, including cafeteria service quality, payment processes, dining practices, and queue management. Pantonial et al. (2025), for instance, examined student payment behaviors, queue management, and cashier efficiency in a higher education institution, while Baguio and Naelga (2024) examined food-service attributes and customer satisfaction in a university cafeteria. These studies emphasize the importance of organized and efficient food-service transactions in educational settings.

In the context of the present study, the researchers' survey revealed concerns regarding students' experiences with the existing manual canteen ordering process. Although 70% of respondents reported that they rarely go to the canteen, 50% stated that they could complete a canteen visit in less than five minutes, while 35% required five to ten minutes. When asked why they were unable to eat their snacks, 59% identified long waiting queues and 55% identified insufficient time. More notably, 92% of respondents reported that they had skipped going to the canteen because of long lines. These findings indicate that waiting time and limited break periods are significant concerns among the respondents. This is consistent with Pantonial et al. (2025), which examined the relationship between payment behavior, queue management, and cashier efficiency, and



with Babasa et al. (2025), which developed an online school-canteen ordering system in response to challenges involving food transactions within limited break periods.

The concern also extends to the management of canteen transactions. When numerous students place orders and make payments at approximately the same time, canteen personnel must process multiple transactions while preparing and distributing food. Cendana and Palaoag (2020) explored the potential of digital payment frameworks for Philippine higher education institutions, while Cabile and Castaño (2025) examined an electronic payment system in a Philippine university in terms of usability, reliability, transaction efficiency, and user satisfaction. These studies provide support for the use of digital technologies to improve transaction processes, although they do not specifically address the pre-ordering and no-show concerns of a campus canteen.

The use of information technology to improve food-service processes is also supported by international research. Breathnach et al. (2020) investigated an online pre-ordering system for a workplace canteen and emphasized the importance of usability and the experience of ordering meals in advance. Migliavada et al. (2021) examined the use of pre-ordering in a university canteen, while Alkhunain et al. (2022) examined online pre-ordering for school lunches. Poovizhi et al. (2025) further explored a digital approach to canteen queue management involving online ordering, queue monitoring, and digital payment. These studies demonstrate that digital technology can be applied to different food-service environments to organize ordering and transaction processes.



Through these functions, **LICEO SAVORE** aims to provide an organized alternative to the existing manual ordering process. Rather than addressing only the placement and payment of orders, the proposed system also considers what happens when a confirmed order is not claimed. The integration of no-show management distinguishes the proposed system from the reviewed solutions and provides a structured approach to managing unclaimed orders. The concerns identified in the study are also connected to the Sustainable Development Goals (SDGs), particularly **SDG 9: Industry, Innovation and Infrastructure** and **SDG 12: Responsible Consumption and Production**. The use of a web-based system to improve the existing manual canteen ordering process aligns with SDG 9, which promotes innovation and the development and adoption of technologies that improve processes and services. Meanwhile, the management of unclaimed pre-orders is related to SDG 12, which promotes responsible resource use and the reduction of waste. By recording and monitoring unclaimed orders, the proposed system can help canteen personnel manage prepared food more systematically and potentially reduce unnecessary food waste.

To address these concerns, this study proposes **LICEO SAVORE: Development of a Web-Based School Canteen Pre-Ordering and Pickup System with Hybrid Payment and No-Show Management to Improve Queue Management and Transaction Efficiency at Liceo de Cagayan University**. The proposed system will allow students to browse the available menu, place food orders before their break, select an available pickup schedule, and choose between Cash on Pickup and online payment through the canteen's designated GCash or Maya QR code. Online payments will be



subject to verification by canteen staff before the order is prepared. The system will also record and monitor unclaimed orders through its no-show management feature.



1.2 Statement of the Problem

Many students buy food at the school canteen during breakfast and break time. However, the canteen currently does not use a technology-driven system for managing food orders, menu availability, and payments. Because of the large number of students, long queues may occur, causing delays for both students and canteen staff.

This study aims to develop a “Campus Canteen Pre-Order System with Hybrid Payment and No-Show Management” to address the following specific problems:

1. The school canteen does not have a technology-driven system for managing food orders, menu availability, and payments.
2. The large number of students buying food during breakfast and break time causes long queues and delays.
3. The manual ordering process makes it difficult for canteen staff to record orders, prepare food, and serve students efficiently.
4. The canteen has no system that allows students to order food in advance and choose between online payment and cash payment at pickup.
5. The acceptability of the proposed system in terms of functionality, usability, reliability, efficiency, and user satisfaction has not yet been determined.



1.3 Objectives of the Study

General Objectives

This study aims to develop and evaluate LICEO SAVORE, a Web-Based School Canteen Pre-Ordering and Pickup System with Hybrid Payment, to improve queue management and transaction efficiency at Liceo de Cagayan University.

Specific Objectives

1. To gather information on the current canteen ordering process and identify the challenges encountered by Grade 12 students and canteen personnel, particularly in terms of waiting time, payment methods, order management, and unclaimed orders.

2. To design a functional requirements of the proposed web-based school canteen pre-ordering and pickup system with hybrid payment based on the identified needs of the users.

3. To develop LICEO SAVORE, a web-based school canteen pre-ordering and pickup system that enables students to place orders in advance, select pickup schedules, choose between cash and online payment, and assists canteen personnel in managing orders, menu availability, payment verification, and no-show records.

4. To evaluate the acceptability of the developed system among Grade 12 students and canteen personnel of Liceo de Cagayan University in terms of:

a. Functionality



- b. Usability
- c. Reliability
- e. User Satisfaction

1.4 Significance of the Study

This study is significant because it addresses common problems in school canteen operations, such as long queues, limited time for students to purchase meals, inefficient order tracking, and financial losses from unclaimed orders. By developing a web-based canteen system with hybrid payment options and QR code integration, the study provides a practical solution that benefits multiple stakeholders. The following groups are expected to benefit from the findings and output of this research:

School. At the institutional level, the system improves overall canteen operations and student service while promoting better meal access for students during school hours.

Future Researchers / Developers. For future researchers and developers, this work serves as a reference for building practical, low-cost school canteen systems. It demonstrates a realistic payment approach using QR codes instead of complex API integration

Students. The system saves students time and avoids long queues during breakfast and break periods, ensuring they can buy food even within limited break times. It also gives them the flexibility to choose between cash or online payment.

Canteen Staff / Administration. For canteen staff and administrators, the system makes it easier to manage and organize incoming orders. It helps estimate how much food to



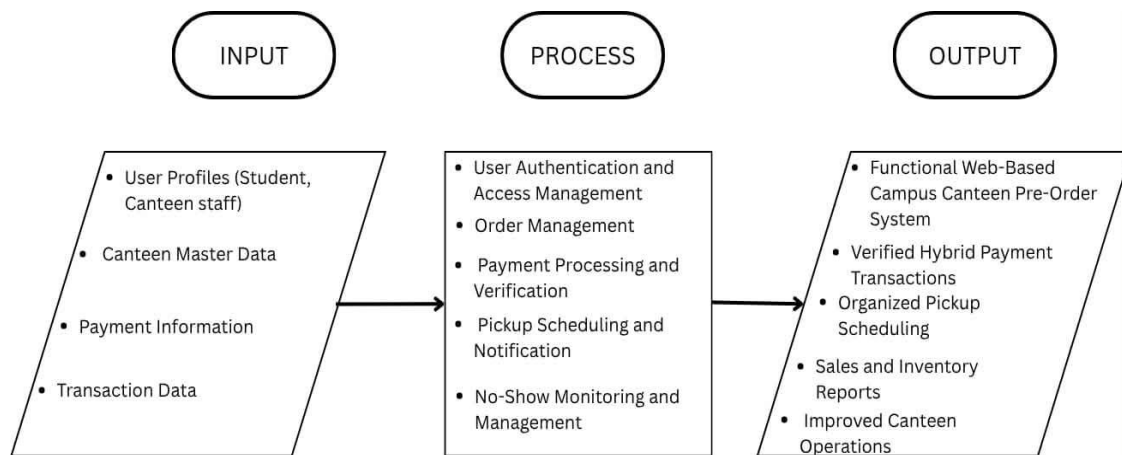
prepare, reducing waste, and reduces financial loss from unclaimed orders through no-show monitoring.

The Future Researchers. For the researchers themselves, the project provides hands-on experience in developing a web-based system with hybrid payment and order management features. It enhances their skills in applying information technology to solve real-world problems in an educational setting.

1.5 Conceptual Framework

Figure 1

Conceptual Framework of the Web-Based School Canteen Ordering and Payment System



This figure presents the Input-Process-Output (IPO) model that guides the development of the proposed system.

The **Input** stage identifies the data and resources needed for the system to operate. This includes user profiles for students and canteen staff, canteen master data such as menu items and pricing, payment information covering both cash and QR code channels, and transaction data from all orders and payments made.

The **Process** stage covers the core operations the system performs. User authentication and access management control what each user type can do. Order management handles the full lifecycle of a canteen order from submission to pickup. Payment processing and verification manages both cash and digital payment channels. Pickup scheduling and notification coordinates order collection and alerts students when their orders are ready. No-show monitoring and management flags unclaimed orders to reduce waste and financial loss.



The **Output** stage presents the expected deliverables. The primary output is a functional web-based campus canteen pre-order system with verified hybrid payment transactions, organized pickup scheduling, and sales and inventory reports that support better canteen operations overall.

1.6 Scope and Delimitation

This study focuses on the development of a Web-Based School Canteen Pre-Ordering and Pickup System with Hybrid Payment for Liceo de Cagayan University – Main Campus. The system is intended for two types of users: Grade 12 students and canteen personnel. It allows students to create an account, browse the available menu, place food orders, select a pickup schedule during morning snack, lunch, or afternoon snack, and choose either Cash on Pickup or GCash/Maya as their payment method. Students can also monitor the status of their orders, cancel orders before food preparation begins, and view their order history and no-show records. On the other hand, canteen personnel can manage menu availability, verify online payments, update order status, monitor no-show records, and generate a daily sales summary. The system also includes a no-show management feature that temporarily suspends students who repeatedly fail to claim their pre-ordered meals, helping improve order management and reduce food waste. To discourage unclaimed orders, students who accumulate three no-shows within a rolling 30-day period are automatically suspended from pre-ordering for seven days, though they may still order in person; the suspension lifts automatically with no manual appeal process.



The system has several limitations. First, it does not perform automatic payment verification, as it is not integrated with a GCash or Maya API; instead of confirming a transaction directly with the payment provider, canteen staff must manually review each uploaded receipt screenshot to confirm that payment was actually made. Second, the system is built exclusively for a single school canteen and is not designed as a multi-vendor platform in the way that services like GrabFood or Foodpanda operate, meaning it cannot support multiple schools or multiple canteens within the same system. Third, because it is a web-based system, it requires an active internet connection — students and canteen staff must either have internet access or be connected to the local school network in order to use it, and the system is not designed to function offline.



1.7 Definition of Terms

LICEO SAVORE – refers to the proposed web-based campus canteen pre-order system designed to organize food pre-ordering, pickup scheduling, hybrid payment, payment verification, and no-show management at Liceo de Cagayan University.

Hybrid Payment – refers to a payment method in the proposed system that allows students to choose between cash payment upon pickup and online payment through the canteen's designated GCash or Maya QR code.

No-Show Management – refers to the process of monitoring and managing orders that are not claimed by students within their selected pickup schedule.

API (Application Programming Interface) – refers to a set of rules and methods that allows different software applications or systems to communicate and exchange data with one another.

Static QR Code – refers to a QR code that contains fixed payment information and does not automatically change for every transaction.

Statistical Process Control (SPC) – refers to a method of using statistical techniques to monitor and evaluate processes to identify variations and determine whether a process is operating consistently.



CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the review of related literature and studies that provide the foundation and support for the development of the proposed LICEO SAVORE: Web-Based School Canteen Pre-Ordering and Pickup System with Hybrid Payment and No-Show Management. It discusses relevant local and international literature and studies concerning digital payment, online food ordering, pre-ordering systems, cafeteria services, queue management, transaction efficiency, and user satisfaction. The reviewed sources provide insights into existing approaches, technologies, and findings related to improving food-service transactions in educational institutions and other similar settings. Furthermore, this chapter identifies the similarities, differences, and limitations of previous studies in relation to the present research. The synthesis of these sources is used to establish the research gap and justify the development of a system that integrates advance food ordering, scheduled pickup, hybrid payment, and no-show management for the Liceo de Cagayan University campus canteen.

2.1 Local Studies and Literature

Digital technology has increasingly been considered as a way to improve transactions and services in Philippine educational institutions. Cendana and Palaoag (2020) examined the potential adoption and acceptance of a digital payment framework for higher education institutions in the Philippines. Their proposed framework explored the use of digital payment technology to improve payment services within educational institutions. The study is relevant to the present research because it demonstrates how digital payment can be incorporated into campus services. However, its focus was primarily on payment services rather than food pre-ordering or canteen queue management.

In relation to students' food-related practices, Trinidad and Balderama (2023) examined the dining practices of college students and the factors associated with their food choices and dining behaviors. The study considered dining practices in relation



to health, sustainability, and food waste, providing an understanding of the factors that influence students' decisions regarding food consumption. This is relevant to the present study because the proposed system involves students' food-ordering activities within a campus environment. It also provides a basis for considering responsible food management as part of a campus food-service system.

Baguio and Naelga (2024) investigated the impact of food-service attributes on customer satisfaction in cafeteria services at a state university in Northern Mindanao. The study included students, teachers, office staff, and other university personnel as respondents and examined how different aspects of cafeteria service relate to customer satisfaction. Its focus on university cafeteria services is relevant to the present research because it demonstrates the importance of service quality in a campus food environment. However, the study focused on cafeteria service attributes and customer satisfaction rather than developing an online pre-ordering system.

The role of electronic payment in improving transactions was further examined by Cabile and Castaño (2025), who evaluated an e-payment system at a private university in Manila. Their study examined system quality in terms of security, reliability, usability, and responsiveness and investigated its relationship with operational efficiency, system effectiveness, and user satisfaction. The researchers reported that system quality was associated with reduced transaction times, lower error rates, and higher user satisfaction, although technical and behavioral barriers could affect the implementation of electronic payment systems. These findings support the inclusion of an electronic-payment option in the proposed canteen system while also emphasizing the importance of usability, reliability, and security.

More specifically related to queuing, Pantonial et al. (2025) investigated student payment behaviors, queue management, and cashier efficiency at Liceo de Cagayan University. The researchers examined transaction times, peak payment periods, digital-payment adoption, queue strategies, and queue monitoring among 761 students. Their findings showed that cashier efficiency was influenced by factors such as payment methods, technology adoption, queue strategies, technology



integration, and queue monitoring. Queue management was also identified as a strong predictor of cashier efficiency. This study is particularly relevant to the present research because it provides evidence from the same university regarding the relationship between payment processes, queue management, and transaction efficiency.

Another local study that directly addresses the use of technology in school canteens is the work of Babasa et al. (2025), who developed COSY, an online ordering web application for school canteens. The system was designed for iACADEMY students and staff to place orders and manage transactions while making better use of limited break periods. The study identified the difficulty of choosing food stalls and completing transactions within a limited amount of time as a problem in the school setting. This study closely relates to the present research because both propose an online ordering approach intended to make food transactions more convenient during school breaks.

Taken together, the local literature and studies demonstrate the potential of digital technology to improve several aspects of campus services, including payment, food-service operations, queue management, and online ordering. Cendana and Palaoag (2020) and Cabile and Castaño (2025) demonstrate the relevance of digital payment in educational institutions, while Baguio and Naelga (2024) and Trinidad and Balderama (2023) provide perspectives on cafeteria services and student dining practices. Pantonial et al. (2025) provides direct evidence concerning queue management and cashier efficiency at Liceo de Cagayan University, while Babasa et al. (2025) demonstrates the application of online ordering in a school canteen. However, these sources do not present a single system that combines advance food ordering, scheduled pickup, cash and online payment, and management of unclaimed orders within the specific context of the Liceo de Cagayan University campus canteen.



2.2 International Studies and Literature

International research has demonstrated that digital pre-ordering can improve the way users interact with food-service systems by making ordering more convenient and organized. In a workplace canteen, users' experiences with an online pre-ordering system showed that the design and usability of the platform influenced how easily employees interacted with the ordering process. Similarly, research involving primary school students found that the presentation of menus and food choices affected how children interacted with online lunch-ordering systems, with differences in independence depending on their age. These findings suggest that digital food-ordering systems should be designed to be simple, understandable, and accessible to their intended users (Breathnach et al., 2020; Alkhunain et al., 2022).

The use of pre-ordering has also been shown to be applicable in educational food-service environments. A three-year study involving 946 university canteen users examined how students adopted and continued to use a pre-ordering system, including the frequency of orders, reservation timing, and the effect of familiarity with the system. The findings indicate that students can adopt digital pre-ordering in university canteens and that continued familiarity with the system may influence its use over time. Similar applications in school food services demonstrate that students can incorporate digital pre-ordering into their regular meal-selection process, indicating its potential for use in educational institutions (Migliavada et al., 2021; Alkhunain et al., 2022).

Beyond basic pre-ordering, digital canteen systems have increasingly incorporated functions intended to improve transaction and queue management. One developed system allowed students to pre-order meals, select pickup times, and make online payments while providing administrators with features for managing orders, updating menus, and monitoring sales. These functions demonstrate how digital ordering can extend beyond simply placing orders by supporting scheduled collection, payment processing, and administrative management. Such features are particularly relevant to canteen environments where long queues and inefficient transaction processes can affect students' overall experience (Poovizhi et al., 2025).



Taken together, these international studies show that digital pre-ordering can support usability, student adoption, and more organized food-service operations across workplace, school, and university settings. The literature emphasizes the importance of user-friendly interfaces, demonstrates the feasibility of pre-ordering among students, and shows the potential of integrating ordering with pickup scheduling and online payment (Breathnach et al., 2020; Migliavada et al., 2021; Alkhunain et al., 2022; Poovizhi et al., 2025). However, the reviewed studies do not specifically combine cash and online payment options with scheduled pickup and no-show management in a Philippine university canteen. This gap provides a basis for developing a system that addresses these functions within the specific context of a Philippine university food-service environment.

2.3 Synthesis of the Related Studies and Literature

The reviewed local and international literature and studies show that technology can support improvements in campus food services through digital payment, queue management, and online food ordering. Local studies have examined digital payment frameworks, cafeteria service quality, student dining practices, queue management, and online canteen ordering. In particular, Pantonial et al. (2025) examined queue management, payment behavior, and cashier efficiency at Liceo de Cagayan University, while Babasa et al. (2025) demonstrated the use of an online ordering system for a school canteen. International studies have likewise explored online pre-ordering in workplace, school, and university food-service environments. These studies provide a foundation for applying digital technology to improve food-ordering and transaction processes.

However, the reviewed sources focus on different aspects of food-service operations and do not present the same combination of functions proposed in the present study. More importantly, none of the reviewed sources incorporates a dedicated no-show management feature for unclaimed pre-orders. Existing studies provide useful approaches to online ordering, payment, queue management, and pre-ordering, but they do not specifically address the problem of students failing to claim their reserved meals after placing an order. This creates a gap in managing the



consequences of unclaimed pre-orders, particularly the possible waste of prepared food and the resulting loss for the canteen.

The proposed Campus Canteen Pre-Order System with Hybrid Payment and No-Show Management addresses this gap by integrating food pre-ordering, scheduled pickup, cash and online payment options, payment verification, and a dedicated no-show management feature. The system will record unclaimed orders and allow canteen staff to monitor repeated no-shows and implement appropriate policies. The no-show management component serves as a distinct feature of the proposed system because it extends the focus of existing digital ordering solutions beyond the placement and processing of orders to also address what happens when an order is not claimed.

Therefore, the proposed system seeks to provide a more comprehensive approach to campus canteen management by addressing both the waiting-time problem before ordering and the management of unclaimed orders after ordering. By combining these functions in one system, the study aims to provide students with a more convenient ordering process while giving canteen staff a structured way to manage orders, payments, pickup schedules, and no-shows.

Table 1.0

Author(s) & Year	Classification	Objectives	Methods	Findings	Gaps
Cendana & Palaog (2020)	Local	Explore digital payment frameworks in Philippine higher education.	Conceptual analysis; policy review and evaluation of existing models.	Digital payment can improve transaction speed, reduce cash handling, and promote campus efficiency.	Digital payment only; no food pre-ordering, pickup scheduling, or no-show management.



Breathnach et al. (2020)	International	Investigate online pre-ordering in a workplace canteen and user experience.	Mixed methods using surveys and interviews.	Online pre-ordering is convenient and can reduce canteen waiting time.	Workplace setting; no hybrid payment or no-show management.
Migliavada et al. (2021)	International	Examine long-term pre-ordering in a university canteen.	Longitudinal study of pre-ordering data over three years.	Pre-ordering improved customer flow, reduced congestion, and improved operations.	No hybrid payment or dedicated no-show management.
Trinidad & Balderama (2023)	Local	Determine college students' dining practices and factors affecting food choices.	Descriptive correlational research using questionnaires.	Convenience, accessibility, available time, and academic schedules influence dining decisions.	Focuses on student behavior, not a digital ordering system.
Baguio & Naelga (2024)	Local	Assess cafeteria service quality and customer satisfaction.	Quantitative research using SERVQUAL and surveys.	Service efficiency and waiting time, among other factors, affect satisfaction.	Service-quality focus rather than an integrated pre-ordering system.
Alkhurain et al. (2022)	International	Evaluate online pre-ordering for school lunches.	Experimental implementation and evaluation.	Advance ordering improved preparation, planning, and efficiency while reducing	No proposed hybrid payment and no-show management features.



				waiting time.	
Cabile & Castaño (2025)	Local	Examine usability, reliability, and efficiency of electronic payment.	Descriptive research and system evaluation.	Electronic payment is convenient and can reduce transaction time.	Electronic payment focus rather than canteen pre-ordering and no-show management.
Pantoni et al. (2025)	Local	Analyze payment behavior, queue management, and cashier efficiency.	Quantitative study using surveys and statistical analysis.	Payment delays contribute to longer queues; efficient payment improves customer flow.	No complete pre-ordering system or dedicated no-show management.
Babasa et al. (2025)	Local	Develop a web-based school canteen ordering system for long lines and limited break time.	SDLC-based system development and user acceptance testing.	Online menu browsing and ordering can help reduce waiting lines in the canteen.	Does not combine hybrid payment, scheduled pickup, and no-show management.



Poovizhi et al. (2025)	International	Design a digital canteen queue-management system.	System development integrating ordering, queue monitoring, and payment.	Online ordering, queue monitoring, notifications , and payment improve operations and reduce congestion.	No dedicated no-show management or the same hybrid-payment setup.
LICEO SAVORE (Present Study)	Proposed Study	Develop a web-based campus canteen pre-ordering and pickup system with hybrid payment and no-show management.	System development using SDLC, testing, and user evaluation.	Students pre-order, select pickup schedules, choose Cash on Pickup or online payment, and track orders; staff manage orders, verify payments, confirm pickup, and monitor no-shows.	Adds dedicated no-show management while integrating pre-ordering, scheduled pickup, hybrid payment, payment verification, and order tracking.



CHAPTER III

RESEARCH METHODOLOGY

This chapter presents the methodology used in conducting the study and developing the proposed LICEO SAVORE: Web-Based School Canteen Pre-Ordering and Pickup System with Hybrid Payment and No-Show Management. It describes the research locale, sampling method, software development life cycle, system architecture, entity relationship diagram, system flowcharts, system requirements, and the current and proposed system processes. The chapter explains how the researchers gathered and considered the needs of the intended users and how these requirements were translated into the design and development of the proposed system. The Agile Software Development Life Cycle is utilized to guide the iterative development, testing, and evaluation of the system. Through these methodological procedures, the researchers aim to develop a system that addresses the identified problems in the existing manual canteen process and supports more organized ordering, payment, pickup, and no-show management.

3.1 Research Locale

This study will be conducted at Liceo de Cagayan University – Main Campus, located at Rodolfo N. Pelaez Boulevard, Kauswagan, Cagayan de Oro City, 9000, Misamis Oriental, Philippines. The university was selected as the research locale because it has a large Senior High School population and an active school canteen that serves students during morning snack, lunch, and afternoon snack breaks. These



peak periods often result in long queues, limited time for purchasing food, and delays in food transactions, making the university an appropriate setting for the study.

The respondents of this study will be Grade 12 Senior High School students enrolled in the Science, Technology, Engineering, and Mathematics (STEM), Humanities and Social Sciences (HUMSS), Accountancy, Business, and Management (ABM), Technical-Vocational-Livelihood Information and Communications Technology (TVL-ICT), and Technical-Vocational-Livelihood Home Economics (TVL-HE) strands. These students were chosen because they regularly use the school canteen during their scheduled breaks and can provide relevant information regarding their experiences with the current food ordering process, waiting time, and overall canteen services.

Liceo de Cagayan University was chosen because the proposed Web-Based School Canteen Pre-Ordering and Pickup System with Hybrid Payment is specifically designed to address the challenges experienced by Grade 12 students when purchasing food during break periods. Based on the researchers' preliminary survey, many students experience difficulties buying food due to long queues and limited break time. Through the proposed system, students will be able to place their food orders in advance, choose either cash or online payment, and pick up their orders during their preferred break schedule. The findings of this study are expected to contribute to improving the efficiency of canteen operations and provide a faster, more convenient, and organized food ordering experience for Grade 12 students and canteen personnel at Liceo de Cagayan University.



3.2 Sampling Method

The study will use purposive sampling to select participants who can provide relevant information about the Campus Canteen Pre-Order System. The respondents will consist of selected Grade 12 students who purchase food from the school canteen and canteen staff who are involved in managing food orders and daily canteen operations.

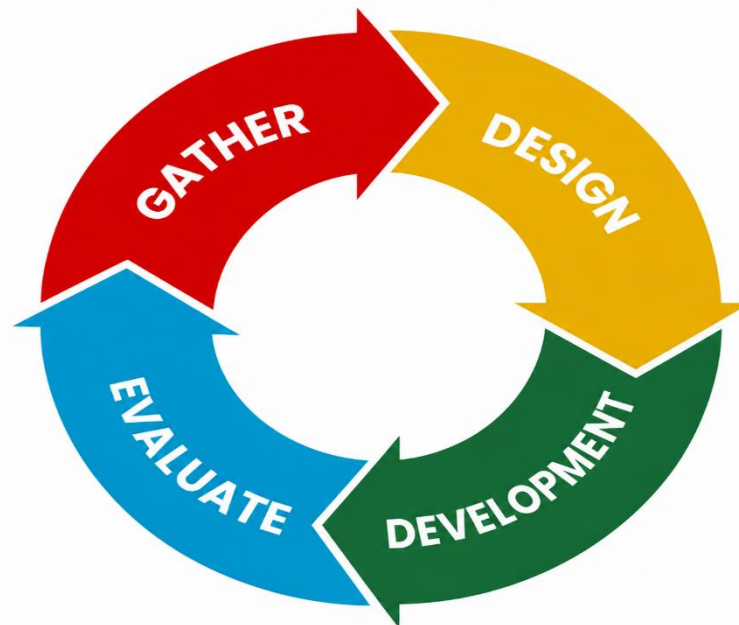
The selected Grade 12 students will be chosen because they have actual experience purchasing food from the school canteen and are therefore able to provide relevant feedback regarding the current ordering process, waiting time, and possible use of a pre-order system. Canteen staff will also be included because they are directly responsible for processing orders and managing daily canteen operations.

This sampling method is appropriate because the participants are selected based on their relevant experience and involvement with the school canteen. Their responses will help the researchers determine whether the proposed Campus Canteen Pre-Order System can improve the ordering process, reduce waiting time, and make canteen services more efficient.

3.3 System Development Life Cycle

Figure 2

Agile



The researchers will adopt the Agile Software Development Life Cycle (SDLC) in developing LICEO SAVORE: A Web-Based School Canteen Pre-Ordering and Pickup System with Hybrid Payment. Agile SDLC is an iterative software development methodology that allows the system to be developed through continuous planning, development, testing, and improvement. The development process will consist of four major phases: Gather, Plan, Develop, and Evaluate.

Phase 1: Gather

During this phase, the researchers will gather information about the existing school canteen ordering process through surveys and consultations with Grade 12 students and canteen personnel. The collected data will be used to identify the



challenges encountered in terms of waiting time, payment methods, order management, and unclaimed orders. The findings will serve as the basis for determining the functional and non-functional requirements of the proposed system.

Phase 2: Design

Based on the gathered information, the researchers will prepare the design and technical requirements of the proposed system. This phase includes designing the system architecture, Entity Relationship Diagram (ERD), Data Flow Diagram (DFD), database structure, user interface, and other design documents that will serve as the blueprint for system development.

Phase 3: Develop

The proposed system will be developed using iterative Agile sprint cycles, with each sprint focusing on a specific module of the system.

Sprint 1 – Database and User Authentication

The researchers will develop the MySQL database and implement the registration and login modules for Grade 12 students and canteen personnel.

Sprint 2 – Menu Management and Pre-Ordering

The researchers will develop the digital menu, shopping cart, food pre-ordering feature, and pickup scheduling for morning snack, lunch, and afternoon snack. Canteen personnel will also be able to manage menu items, prices, and food availability.

Sprint 3 – Hybrid Payment



The checkout module will be developed to allow students to choose between Cash on Pickup and GCash/Maya payment. Students who select online payment will upload their payment receipt for manual verification by the canteen personnel.

Sprint 4 – Order Management and No-Show Monitoring

The researchers will develop the administrative dashboard for monitoring incoming orders, verifying payments, updating order status, managing menu availability, generating daily sales summaries, and recording students with unclaimed orders through the no-show management feature.

After each sprint, the completed module will undergo testing to identify and correct errors before proceeding to the next sprint. This iterative process ensures that every module functions properly before integration into the complete system.

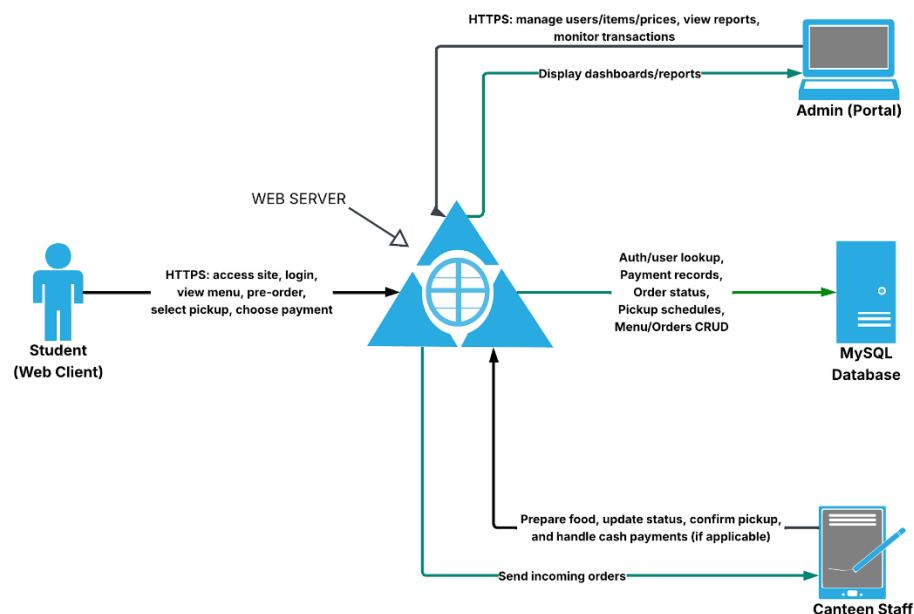
Phase 4: Evaluate

After the development of the proposed system, it will be evaluated by selected Grade 12 students and canteen personnel of Liceo de Cagayan University. The evaluation will determine the level of acceptability of the system in terms of Functionality, Usability, Reliability, Efficiency, and User Satisfaction. The results of the evaluation will be used as the basis for identifying improvements and determining whether the proposed system achieved its intended objectives.

3.4 System Architecture

Figure 3

Overall structure of the system — components, modules, and how they interact



The System Architecture Diagram illustrates how the proposed Campus Canteen Pre-Order System with Hybrid Payment and No-Show Management operates by connecting the student, web server, database, canteen staff, and administrator. It shows the flow of information between each component to ensure efficient order processing, payment management, and system administration.

The process begins with the Student (Web Client), who accesses the system through a secure HTTPS connection using a web browser. The student logs into the system, views the available menu, places a pre-order, selects a preferred pickup schedule, and chooses a payment method, either online or cash on pickup. All user requests are sent to the Web Server for processing.



The Web Server serves as the central component of the system. It authenticates user accounts, processes orders, manages payment information, updates order status, and communicates with the MySQL Database. The database securely stores essential information, including user accounts, food menu items, orders, payment records, pickup schedules, and transaction history. This ensures that all data is organized, accurate, and available whenever needed.

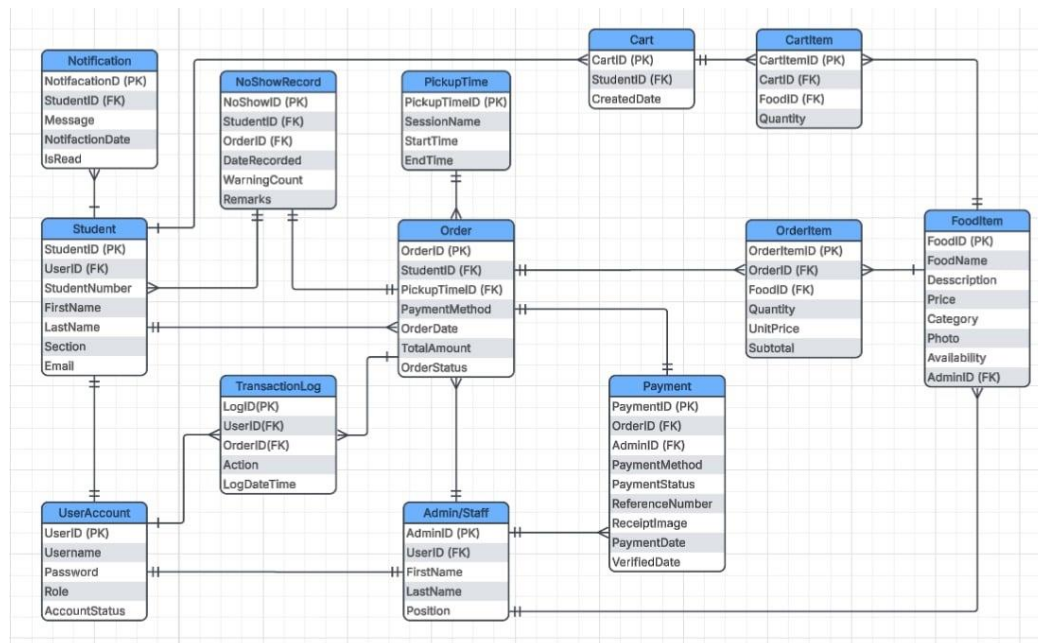
For the Canteen Staff access the system through the web server to receive incoming orders from students. They prepare the requested food, update the order status, confirm when orders are ready for pickup, and process cash payments for students who selected the cash-on-pickup option. Their updates are saved in the database and are immediately reflected in the system.

Lastly, Administrator (Admin Portal) also connects to the web server through a secure HTTPS connection. The administrator manages users, food items, menu prices, and system settings while monitoring transactions and generating reports through dashboards. All administrative actions are stored in the database to maintain data integrity and support efficient system management.

3.5 Entity Relationship Diagram

Figure 4

Database Structure of the Program



The Entity Relationship Diagram (ERD) illustrates the database structure of the proposed Campus Canteen Pre-Order System with Hybrid Payment and No-Show Management. It shows how different entities are connected to manage user accounts, food ordering, payments, notifications, and no-show records.

The UserAccount entity stores the login credentials and account information of all system users. Each account is assigned a role, which determines whether the user is a Student or an Admin/Staff. The Student entity contains the student's personal information, while the Admin/Staff entity stores the details of canteen personnel responsible for managing food items, orders, and payment verification.

Students can browse available menu items listed in the FoodItem entity and temporarily save their selections in the Cart. Each selected item is stored in the



CartItem entity, allowing students to review or modify their choices before placing an order.

Once the student confirms the purchase, the system creates an entry in the Order entity. Each order is linked to a selected PickupTime schedule and contains details such as the payment method, total amount, order date, and current order status. The specific food items included in the order are recorded in the OrderItem entity, which stores the quantity, unit price, and subtotal for each item.

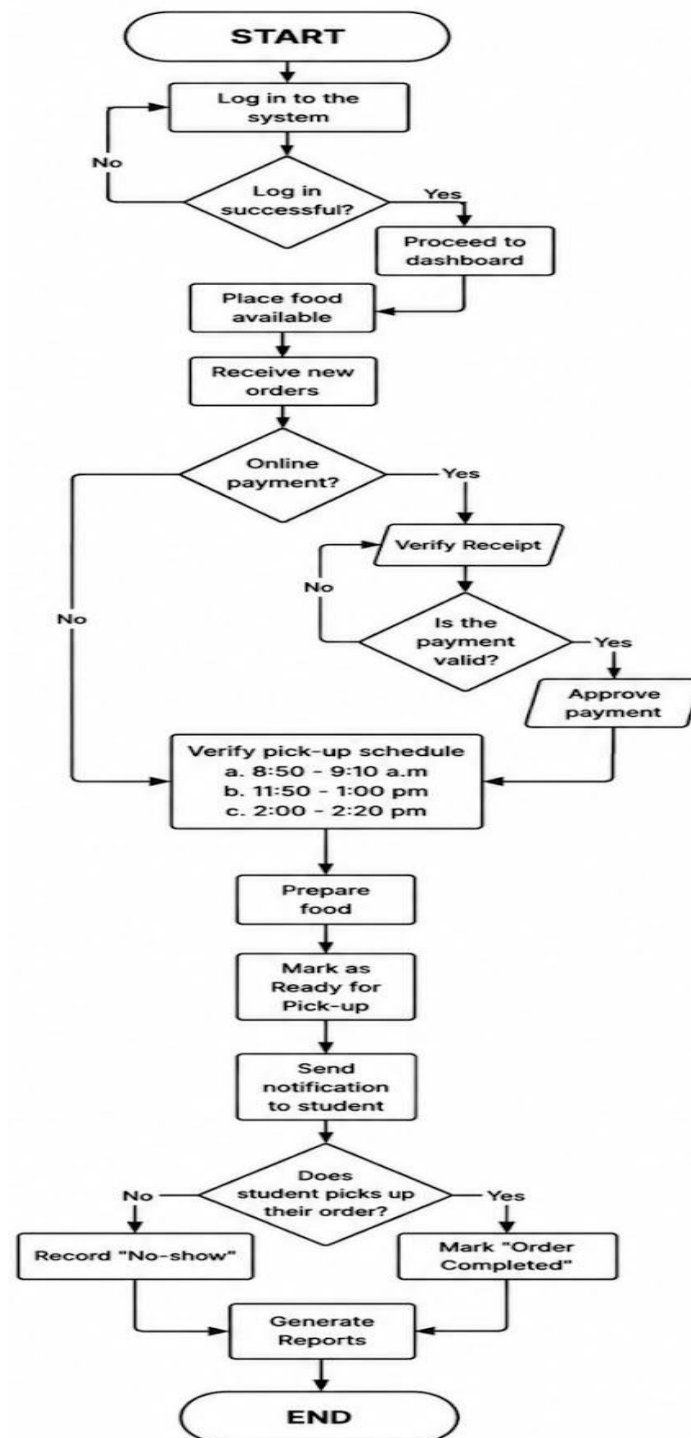
The Payment entity records all payment transactions associated with an order. It stores important information such as the payment method, payment status, reference number, receipt image for online payments, payment date, and verification date. This enables the system to support both online payments and cash-on-pickup transactions.

To improve communication with users, the Notification entity sends updates to students regarding order confirmations, payment verification, food readiness, and other important announcements. Meanwhile, the NoShowRecord entity keeps track of students who fail to claim their pre-orders. It records the warning count, date of occurrence, and remarks, allowing the system to implement no-show management policies.

3.6 Student Flowchart

Figure 5.1

System Flowchart for the Student



3.7 Admin Flowchart

Figure 5.2

System Flowchart for the Administrator

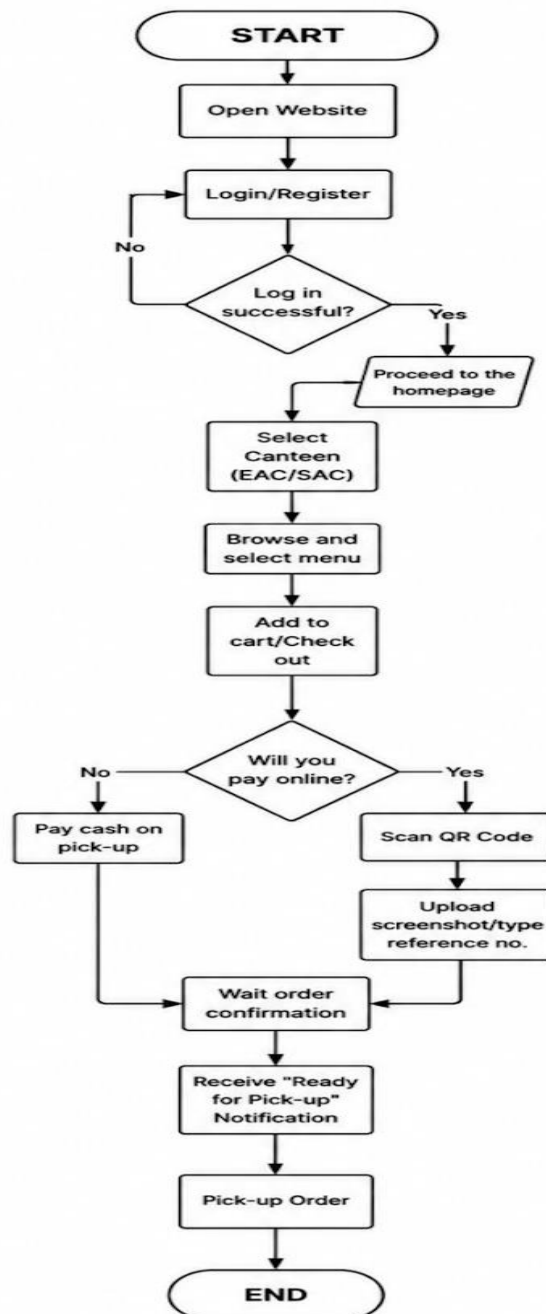




Figure 5.1 shows the workflow of the student in using the Campus Canteen Pre-Order System. The process begins when the student logs into the system using their account credentials. After successfully logging in, the student accesses the menu, browses the available food items, and selects their preferred meals. The student then chooses a pick-up schedule and proceeds to checkout.

At checkout, the student selects a payment method. If online payment is chosen, the student uploads the payment receipt for verification. Once the payment is approved, or if another payment method is selected, the order is successfully placed. The student then receives updates regarding the order status. When the canteen staff marks the order as **Ready for Pick-up**, the system sends a notification to the student. Finally, the student claims the order at the selected pick-up time, completing the transaction. This process ensures a faster, more organized, and convenient food ordering experience while minimizing waiting time at the school canteen.

Figure 5.2 illustrates the workflow of the administrator in managing the Campus Canteen Pre-Order System. The process begins when the administrator logs into the system. If the login credentials are incorrect, the administrator is prompted to log in again. Once the login is successful, the administrator accesses the dashboard, updates the list of available food items, and receives new customer orders.

When an order is received, the administrator checks whether the customer selected online payment. If online payment is chosen, the administrator verifies the uploaded payment receipt. If the payment is valid, it is approved; otherwise, the receipt must be verified again until a valid payment is confirmed. If another payment method is selected, the system proceeds directly to the next step without payment



verification. The administrator then verifies the student's selected pick-up schedule, prepares the ordered food, marks it as **Ready for Pick-up**, and sends a notification to the student. Finally, the administrator checks whether the student has collected the order. If the order is picked up, it is marked as **Order Completed**. Otherwise, it is recorded as a **No-show**. The system then generates reports of completed and unclaimed orders before ending the process.

3.8 System Requirements

Table 2.0

User Requirements

User Requirements	Minimum Requirements
Web Browser	Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari
Internet Connection	Stable connection (minimum 5 Mbps recommended)
Device	Desktop/Laptop/Tablet/Smartphone
Operating System	Windows, macOS, Android, or iOS

Developmental Requirements

Developmental Requirements	Minimum Requirements
Operating System	Windows 10/11
Web Server	Apache (XAMPP)
Database	MySQL
Programming Language	PHP
Code Editor	Visual Studio Code
Markup Language	HTML
Style Sheet	CSS
Browser	Google Chrome, Mozilla Firefox, Microsoft Edge
Processor	Dual-core Intel Core i3 or equivalent
Memory (RAM)	Minimum 4 GB
Storage	Minimum 10 GB available
Security	SSL/TLS (HTTPS) recommended

The User Requirements specify the minimum resources needed by students and canteen staff to access and use the LICEO SAVORE web-based system. The system can be accessed through common web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari, allowing users to access the website using different devices. A stable internet connection of at least 5 Mbps is recommended to support smooth access to the website and its online features. The system can be used through a desktop, laptop, tablet, or smartphone, making it accessible across different devices. It supports commonly used operating systems, including Windows, macOS, Android, and iOS.

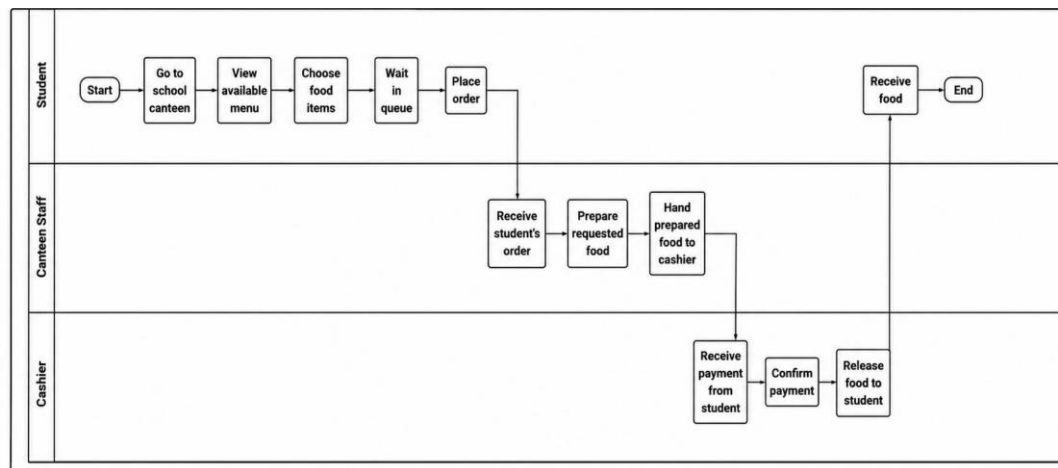


The Developmental Requirements specify the hardware, software, and development tools required to create and run the LICEO SAVORE system during development. The development environment requires Windows 10 or 11, with Apache through XAMPP serving as the local web server and MySQL serving as the database for storing system information such as user accounts, menus, orders, payments, and no-show records. PHP is used as the primary programming language for the server-side functions, while HTML structures the web pages and CSS manages their appearance. Visual Studio Code is used as the code editor for developing the system. A supported web browser is required for testing and accessing the website during development. The development computer should have at least a dual-core Intel Core i3 or equivalent processor, 4 GB of RAM, and 10 GB of available storage. Lastly, SSL/TLS (HTTPS) is recommended to provide a more secure connection when the system is deployed online.

3.9 Current System Flow

Figure 6

Swimlane Diagram



The current canteen system uses a manual food-ordering process. Students first approach the canteen and wait in line until it is their turn to place an order. They inform the canteen staff of their chosen food items, and the staff prepares the order based on the available food and stock. After the order is prepared, the student proceeds with the payment and receives the purchased food.

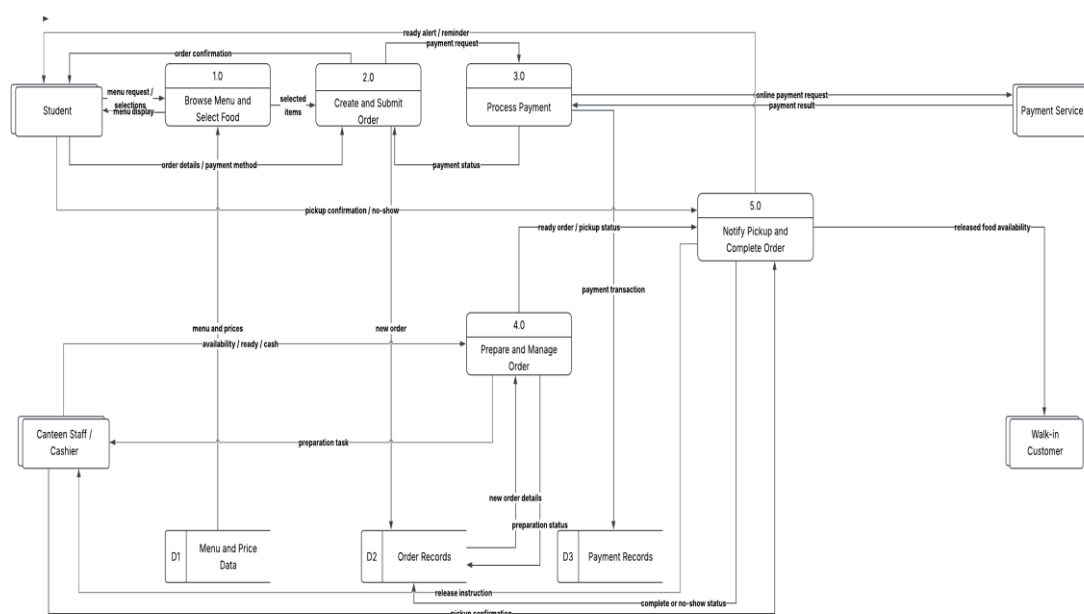
This process requires students to physically line up and wait before their orders can be processed. During breakfast and break periods, when many students purchase food at the same time, the manual process can result in longer queues, increased waiting time, and congestion around the canteen. Canteen staff must also handle orders and payments simultaneously, which can make transactions more difficult to manage during peak periods.

Since there is currently no pre-ordering system on campus, students cannot place their orders in advance or reserve a pickup schedule. They must complete the ordering

process at the canteen itself. This manual process is the main concern that LICEO SAVORE aims to address by introducing online pre-ordering, scheduled pickup, hybrid payment, and no-show management

3.10 Proposed System Flow

Figure 7



The *Gane Sarson* diagram illustrates the overall workflow of the Campus Canteen Pre-Order System by identifying the roles and interactions of the **Student**, **Canteen Staff/Cashier**, and the **System** throughout the ordering process. It clearly defines the responsibilities of each participant and demonstrates how information and tasks move between them to ensure a smooth and organized transaction.

The process begins when the student accesses the website and logs into the system. After logging in, the student browses the available menu, selects the desired food items, chooses a preferred pick-up schedule, and proceeds to checkout. During



checkout, the student selects either **Online Payment** or **Cash on Pickup** before confirming the order.

Once the order is submitted, the system records the transaction and begins processing it. If the student chooses online payment, the system verifies the payment details and sends an order confirmation once the payment is successfully processed. Throughout the preparation stage, the system keeps track of the order status and automatically notifies the student when the food is ready for collection.

At the same time, the canteen staff or cashier receives the order details through the system and prepares the requested items according to the selected pick-up schedule. After the order has been completed, the staff updates its status to **Ready for Pickup**, allowing the system to send a notification to the student. If the student selected **Cash on Pickup**, the cashier collects the payment when the order is claimed.

The final stage of the process determines whether the student successfully picks up the order. If the order is collected, the transaction is marked as completed. However, if the student fails to claim the order within the designated pick-up period, the system records it as a **No-Show Report**, enabling the canteen management to monitor unclaimed orders and improve operational planning.



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APPENDIX A

SURVEY QUESTIONNAIRE

Campus Canteen Pre-Order System

A research instrument for gathering data on
canteen pre-ordering needs and acceptance

Liceo de Cagayan Univeristy

Group 1 - ICT 1

QUESTIONNAIRE 1: STUDENT SURVEY

Instructions: Read each question carefully. Check (✓) the box that corresponds to your answer. For open-ended questions, write your answer on the space provided.

PART 1: DEMOGRAPHIC PROFILE

Name:

Grade Level:

☐ Grade 11

☐ Grade 12

Gender:

☐ Male

☐ Female

☐ Prefer not to say



How often do you eat at the school canteen?

- ☐ Every day
- ☐ 3-4 times a week
- ☐ 1-2 times a week
- ☐ Rarely

PART 2: CURRENT CANTEEN EXPERIENCE

How long do you usually wait in line at the canteen during break time?

- ☐ Less than 5 minutes
- ☐ 5-10 minutes
- ☐ 10-20 minutes
- ☐ More than 20 minutes

What is the biggest problem you experience at the canteen? (Check all that apply)

- ☐ Long waiting time
- ☐ Food runs out before I can order
- ☐ I don't know what food is available
- ☐ The line is confusing
- ☐ I don't have enough time to eat during break
- ☐ Other: _____

Have you ever skipped eating at the canteen because of the long line?

- ☐ Yes
- ☐ No



If yes, what did you do instead?

- ☐ Bought food from outside the school
- ☐ Brought food from home
- ☐ Did not eat at all
- ☐ Other: _____

How do you currently check what food is available at the canteen?

- ☐ I go to the canteen and look
- ☐ I ask friends
- ☐ There is a posted menu
- ☐ Other: _____

PART 3: PRE-ORDER SYSTEM ACCEPTANCE

Would you use a website or app to pre-order food before break time?

- ☐ Yes, definitely
- ☐ Maybe
- ☐ No

What features would you want in a pre-order system? (Check all that apply)

- ☐ See the menu with prices and photos
- ☐ See what food is currently available
- ☐ Order and pay online (GCash/Maya)
- ☐ Order online but pay cash when I pick up



- ☐ Choose a specific pickup time
- ☐ Get a notification when my order is ready
- ☐ Cancel my order if I change my mind
- ☐ Rate the food after eating
- ☐ Other: _____

How far in advance would you want to place your order?

- ☐ The night before
- ☐ In the morning before break
- ☐ 30 minutes before break
- ☐ Anytime, as long as I can pick it up during break

Would you prefer to pay online or pay cash when picking up?

- ☐ Online payment (GCash/Maya)
- ☐ Cash on pickup
- ☐ Both options available

PART 4: NO-SHOW SCENARIO

If you pre-order food and pay cash, but forget to pick it up, what should happen?

- ☐ The food should be given to someone else
- ☐ I should still be charged for it
- ☐ I should get a warning but no charge
- ☐ Other: _____

What would remind you to pick up your order? (Check all that apply)



- ☐ SMS notification
- ☐ App notification
- ☐ A bell or announcement in class
- ☐ My friends reminding me
- ☐ Other: _____

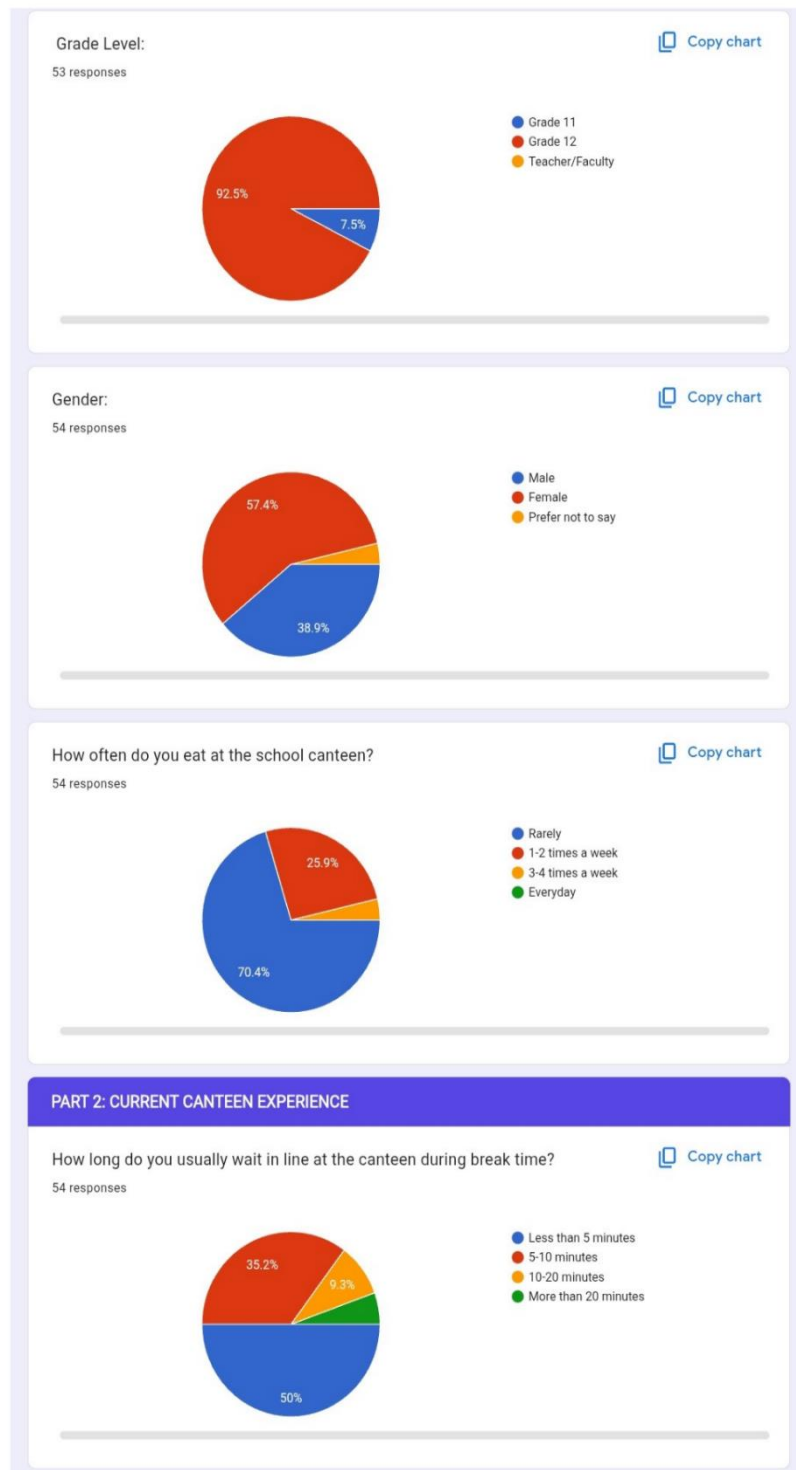
Would you be willing to pay a small deposit (e.g., 10-20 pesos) for pre-orders to ensure you pick up?

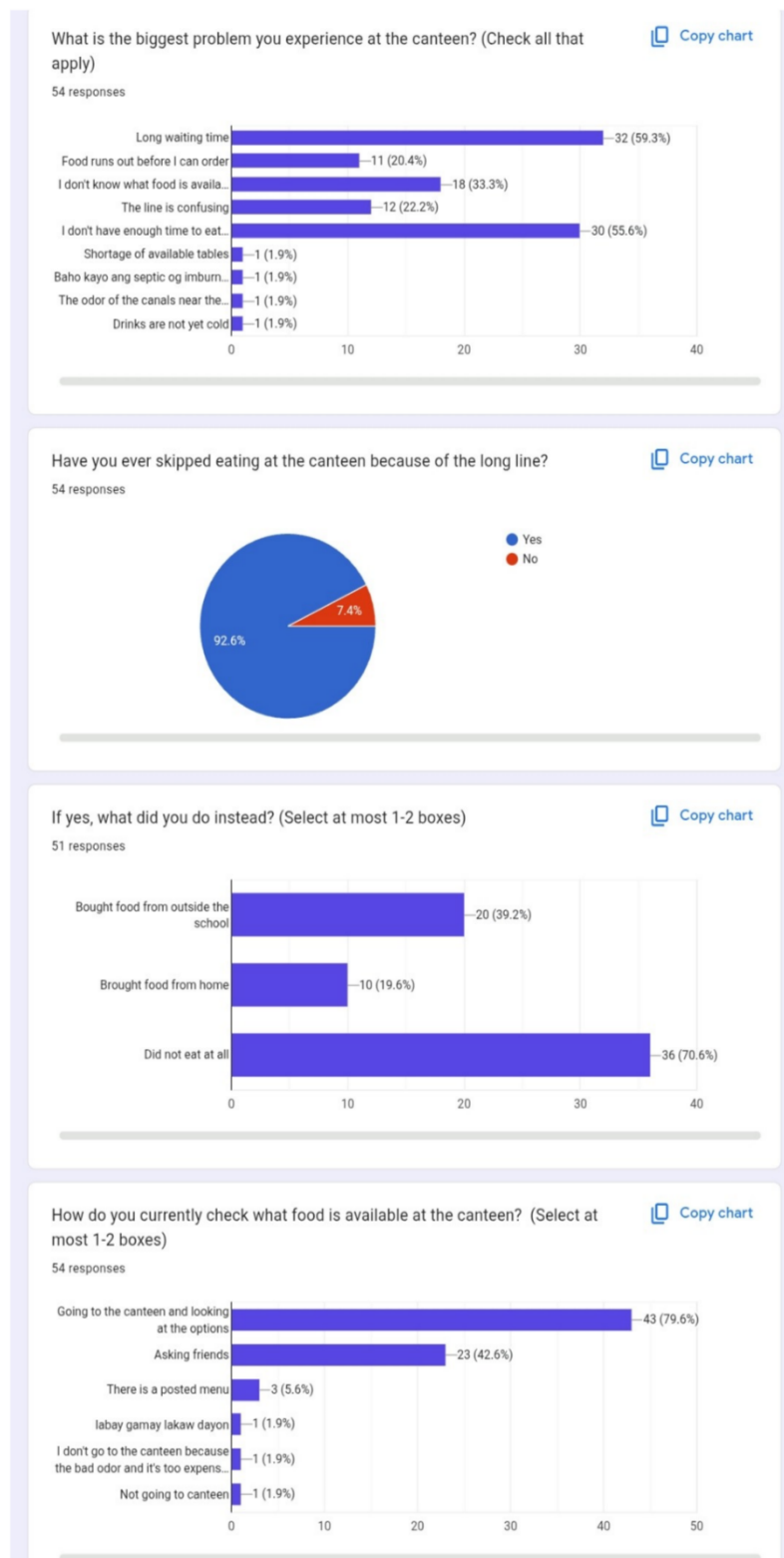
- ☐ Yes
- ☐ No
- ☐ Maybe

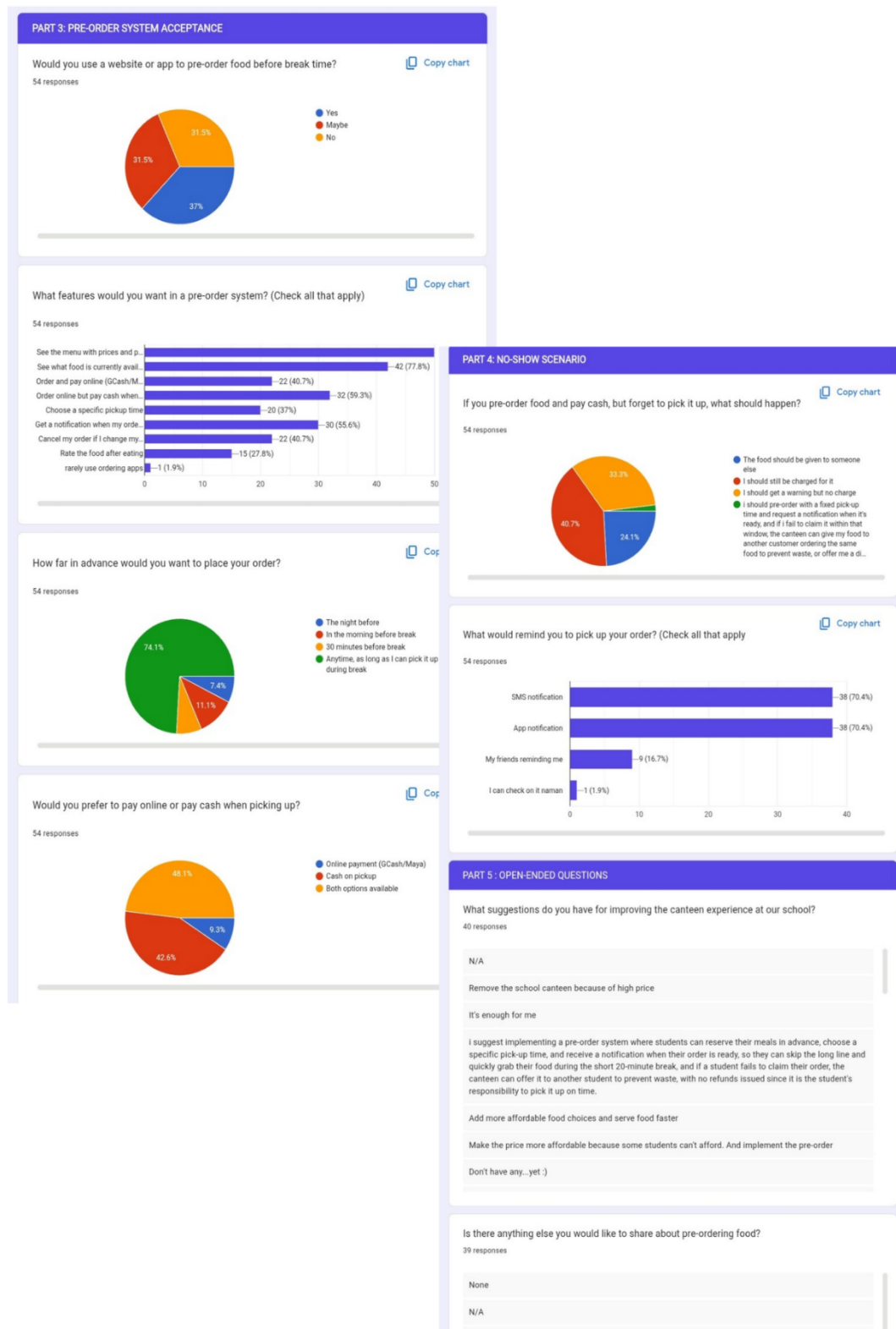
PART 5: OPEN-ENDED QUESTIONS

What suggestions do you have for improving the canteen experience at our school?

Is there anything else you would like to share about pre-ordering food?







QUESTIONNAIRE 2: CANTEEN STAFF SURVEY

QUESTIONNAIRE 2: CANTEEN STAFF SURVEY
Campus Canteen Pre-Order System

Instructions: Read each question carefully. Check (✓) the box that corresponds to your answer. For open-ended questions, write your answer on the space provided.

PART A: BACKGROUND

Name: _____

1. How long have you worked at the school canteen?

☐ Less than 1 year

☒ 1-3 years

☐ More than 3 years

2. What are your main responsibilities? (Check all that apply)

☒ Taking orders

☒ Preparing food

☐ Handling payments

☐ Managing inventory

☐ Other: _____

PART B: CURRENT PROCESS

3. How do students currently place their orders?

☐ They line up and order at the counter

☒ They shout their orders

☐ They send someone to order for them

☐ Other: _____

4. What is the most challenging part of managing orders during break time?

☒ Too many orders at once

☐ Students changing their minds

☐ Not knowing how much food to prepare

☐ Handling cash payments

☐ Other: _____

5. How do you decide how much food to prepare each day?

☒ Based on experience

☐ Based on yesterday's sales

☐ I prepare a fixed amount

☐ Other: _____

PART C: SYSTEM ACCEPTANCE

6. Would you be willing to use a mobile phone or tablet to manage orders?

☒ Yes

☐ Maybe, with training

☐ No

7. What features would help you most? (Check all that apply)

☐ A screen showing all current orders

☒ A way to update available food in real time

☐ A summary of how many orders per food item

☐ A way to mark orders as "ready for pickup"

☐ Other: _____

8. What concerns do you have about using a digital ordering system?

☒ I am not comfortable with technology

☐ It might be too complicated

☐ It might break down

☐ I prefer the current way

☐ No concerns

☐ Other: _____

9. What suggestions do you have for improving the canteen ordering process?

— For Portal

—

—

10. Is there anything else you would like to share about managing canteen orders?

— Maika makasabay ang students

— makay madagayan.

—

QUESTIONNAIRE 2: CANTEEN STAFF SURVEY
Campus Canteen Pre-Order System

Instructions: Read each question carefully. Check (✓) the box that corresponds to your answer. For open-ended questions, write your answer on the space provided.

PART A: BACKGROUND

Name: _____

1. 1. How long have you worked at the school canteen?

☐ Less than 1 year
☒ 1-3 years
☐ More than 3 years

2. 2. What are your main responsibilities? (Check all that apply)

☒ Taking orders
☒ Preparing food
☐ Handling payments
☐ Managing inventory
☐ Other: _____

PART B: CURRENT PROCESS

3. 3. How do students currently place their orders?

☐ They line up and order at the counter
☒ They shout their orders
☐ They send someone to order for them
☐ Other: _____

4. 4. What is the most challenging part of managing orders during break time?

☒ Too many orders at once
☐ Students changing their minds
☐ Not knowing how much food to prepare
☐ Handling cash payments
☐ Other: _____

5. 5. How do you decide how much food to prepare each day?

☒ Based on experience
☒ Based on yesterday's sales
☐ I prepare a fixed amount
☐ Other: _____

PART C: SYSTEM ACCEPTANCE

6. 6. Would you be willing to use a mobile phone or tablet to manage orders?

☒ Yes
☐ Maybe, with training
☐ No

7. 7. What features would help you most? (Check all that apply)

☐ A screen showing all current orders
☐ A way to update available food in real time
☒ A summary of how many orders per food item
☐ A way to mark orders as "ready for pickup"

8. 8. What concerns do you have about using a digital ordering system?

☐ I am not comfortable with technology
☐ It might be too complicated
☐ It might break down
☒ I prefer the current way
☐ No concerns
☐ Other: _____

9. 9. What suggestions do you have for improving the canteen ordering process?

pos

10. 10. Is there anything else you would like to share about managing canteen orders?

unta taas lang postura than naman.
wit kay tao rason ni nasyop.



APPENDIX B

AI USE DECLARATION AND CONTEXTUALIZATION

Research Section	AI Tool Used	Purpose / Type of Assistance	Researcher's Contextualization, Verification, and Refinement
Background of the study	Anara	Helped us gather ideas and information that could be used in writing the background of the study.	We reviewed the information, checked if it was related to our topic, and revised it to fit our research.
Review of Related Literature	Concensus	Helped us search for relevant academic sources related to our research topic.	We checked the sources for relevance and credibility before using and properly formatting them in our research.
	Anara	Summarizing key findings	We checked the information and sources, then edited and organized the content based on our research variables and objectives.
Sampling Method	ChatGpt 5.0	Helped us understand and choose the appropriate sampling method for our study.	We considered the suggestions and adjusted them based on our respondents, population, and research design.



Liceo de Cagayan University - Senior High School Department

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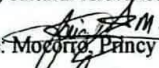
System Requirements	Claude	Helped us identify and organize the functional and non-functional requirements of our system.	Our group reviewed the suggestions and modified them according to the actual features and needs of our proposed system.
System Architecture	Claude	Give us ideas on how to organize about the components and structure of our system architecture.	Our group understands, reviewed, and adjusted the information to make sure it accurately represented how our proposed system work.
UI/UX Design	Google Stitch	Gives prompt for our website.	The member checked the prompt and changed something that is not necessary for our system

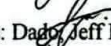
Declaration Statement

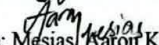
I, the undersigned, declare that Artificial Intelligence (AI) tools were used only to assist in language, structure, and organizational suggestions throughout the writing of this research.

The generation of codes, categories, memos, and theory construction were entirely conducted by the researcher following grounded theory protocols.

All AI-assisted outputs were reviewed, contextualized, and reinterpreted to ensure alignment with actual data, cultural relevance, and academic integrity.

Researcher's Name and Signature:  Mojonio, Princy Laiza

Researcher's Name and Signature:  Dado, Jeff Brian

Researcher's Name and Signature:  Mesias, Aaron Khalil



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Page

Researcher's Name and Signature: Responso, Andrew Alessy

Researcher's Name and Signature: Gualiza, Christian Rhey

Researcher's Name and Signature: Vetez, Jandy

Researcher's Name and Signature: Ralloma, Mark Clarence

Researcher's Name and Signature: Cangit, Dennis

Researcher's Name and Signature: Loquias, Ken

Researcher's Name and Signature: Aerda, Rogel

Researcher's Name and Signature: Albiso, Mary Grace

Researcher's Name and Signature: Jumali, John Dave

Date: September 9, 2026

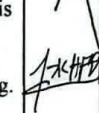
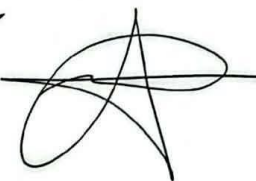
Adviser's Name and Signature: Izunian Thaise Diaz



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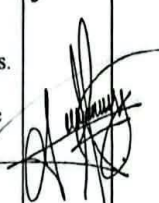
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APPENDIX C

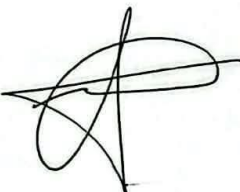
Date	Topics Discussed	Signature	
		Advisee	Adviser
AUGUST 14, 2026	We let our research adviser check our chapter 1 to 3. Mostly to see if our background, rri, etc., is correct. Our research adviser told us that some parts need fixing. We just noted the comments so we can edit it after.	   	



Absent			

Date	Topics Discussed	Signature	
		Advisee	Adviser
AUGUST 25, 2026	We showed our ppt, wireframe, and diagrams. Our adviser looked at the design and layout, said that there's something wrong with our diagram. She told us to make it simple and not too colorful. We listen and changed it after.	  Ann Pascas. Li. Migz. Joni.	



			
Liceo de Cagayan University - Senior High School Department			Page
			
Absent	Jumahali, Albiso, Acido, Velez, Caingin, Loquias		